

Maestro Cloud Control

Enabling Access to AWS Billing and Management

Admin Guide

M3AG-2

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Version 3.2

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PREFACE

ABOUT THIS GUIDE

The guide gives detailed instructions on what steps should be taken to enable programmatic access to AWS Account billing and how to allow managing virtual resources.

These steps include enabling billing data collection and enabling management .

AUDIENCE

The guide is designed for administrators of AWS accounts, who want to enable access to its billing information and account management for Maestro application.

PRE-REQUISITES

Before you can perform the steps described in the guide, ensure that these conditions are met:

1. You have an active AWS account.
2. You have access to the account with root user permissions.
3. You know the ID of the account to which the access will be provided.

EXPECTED RESULTS

Once you have performed all the steps described in the guide, you will get this outcome:

1. A Cost and Usage report created and configured for your account.
2. An IAM role for enabling access to billing data of your account.
3. An IAM role for enabling management of infrastructure on your account.
4. A file or several files which contain this data for further sharing with Maestro3 team:
 - 4.1. For billing:
 - 4.1.1.The ID of the AWS Account from which billing will be gathered
 - 4.1.2.Athena region code (for example, eu-central-1). See this page to check it. • Cost and Usage reports S3 bucket name
 - 4.1.3.Athena Query result S3 bucket name
 - 4.1.4.Billing Access Role ARN name
 - 4.1.5.Glue Database name
 - 4.1.6.Glue Table name
 - 4.1.7.Cost and Usage reports status file S3 Path
 - 4.2. For management:
 - 4.2.1.The ID of the AWS Account for which management via Maestro should be enabled.
 - 4.2.2. The .csv file with the secrets for programmatic access

1 INTRODUCTION

To connect your AWS accounts to Maestro, you need two type of operations – **gathering billing data** and **managing virtual infrastructures**. This document gives step-by-step instructions for both these processes.

If you have only one account in AWS, both enabling billing and management access instructions should be performed on it. This rule also applies when you have several independent accounts in AWS: If this is the case, follow both instructions for each account you want to connect to Maestro.

If you use AWS Consolidated Billing, apply the Enabling Billing Data collection instructions to **Master account only**, and apply the Enabling Management instructions to all linked accounts for which you want to enable the infrastructure management via Maestro.

2 ENABLING BILLING DATA CONNECTION

Maestro collects billing data from the connected AWS accounts, processes it, and then delivers to users as reports on UI and via emails.

To enable these operations, you need to do this:

1. Create a Cost & Usage Report (CUR 2.0 – recommended, or CUR)
2. Configure the AWS Athena service.
3. Create an IAM role for enabling billing access.
4. Make two additional steps to retrieve information for sharing.



If you use AWS Consolidated Billing, apply the instructions below to the Master account.

All resources used to gather billing data in AWS should be created in one region.

In the guide, we use eu-west-1 (Ireland), as it is the region where Maestro is hosted.

2.1 CREATING A COST AND USAGE REPORT

Enabling billing data is possible via two types of AWS Cost and Usage reports – CUR (the former one) and the latest one, CUR 2.0 (supported after it was introduced on AWS side). Using CUR 2.0 is recommended as it is a more optimized and recent approach. However, CUR is also possible, especially if you already use it for your account.

According to AWS, the CUR 2.0 table provides the same information as Cost and Usage Reports (CUR) with a few improvements:

- Cost and Usage Reports 2.0 provides the following improvements over Cost and Usage Reports:
- Consistent schema: CUR 2.0 contains a fixed set of columns, whereas the columns included for CUR can vary monthly depending on your usage of AWS services, cost categories, and resource tags.
- Nested data: CUR 2.0 reduces data sparsity by collapsing certain columns from CUR into individual columns with key-value pairs of the collapsed columns. The nested keys can optionally be queried in Data Exports as separate columns to match the original CUR schema and data.
- Additional columns: CUR 2.0 contains two additional columns: `bill_payer_account_name` and `line_item_usage_account_name`.

You can find more details on AWS CUR 2.0 in the [official documentation](#).

You can find the details on creating each type of the report further in this section.

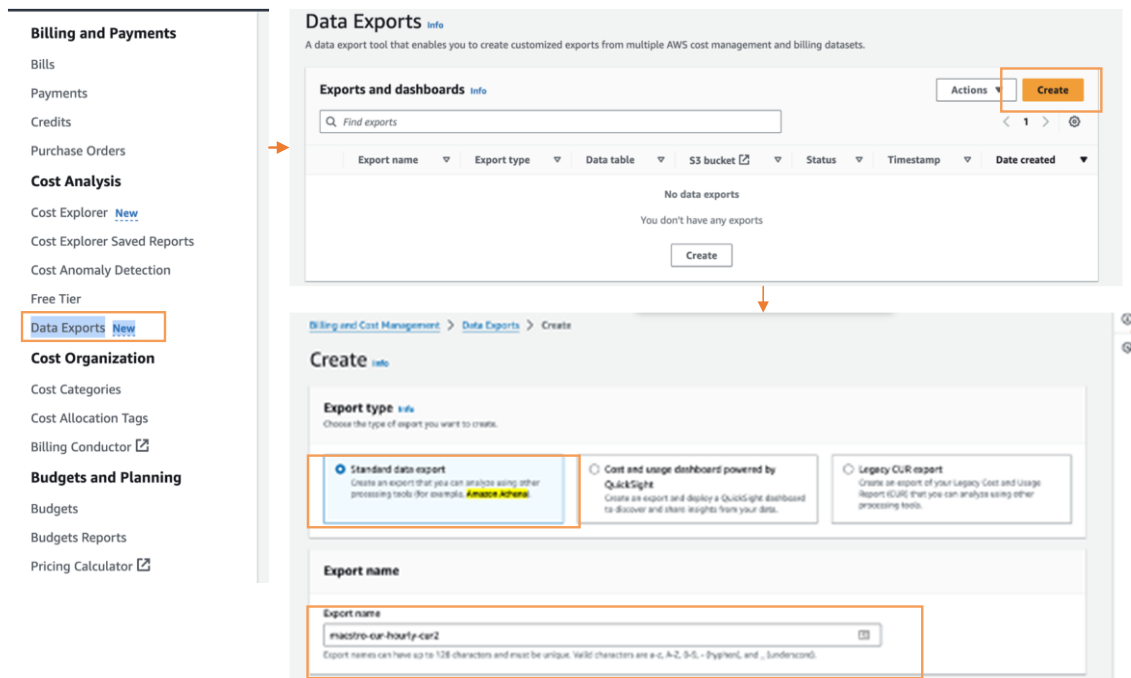
2.1.1 Creating a CUR 2.0 Report

Using CUR 2.0 report type is recommended due to its better performance and optimization.

2.1.1.1 Configuring report

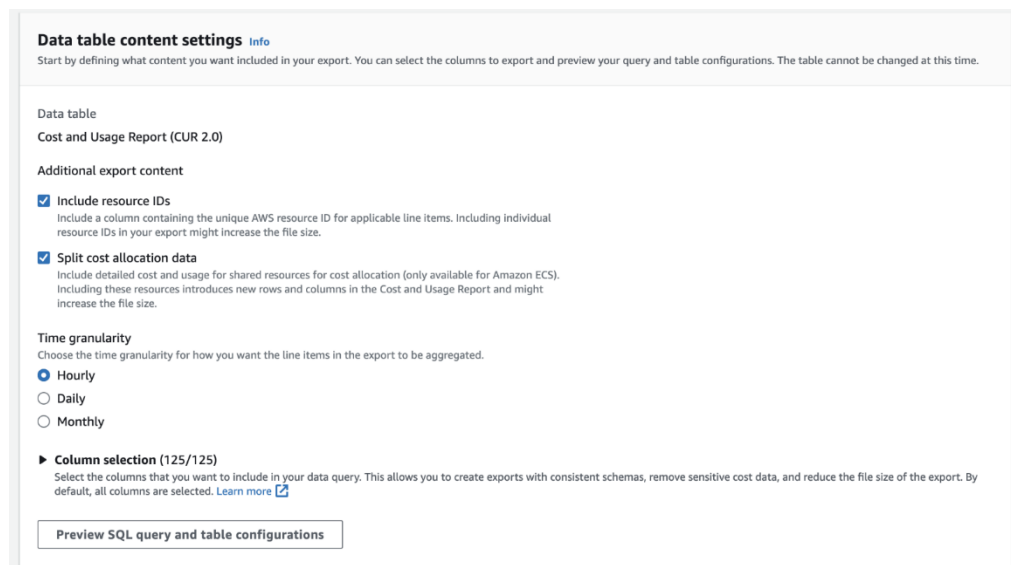
To configure the report, do this:

1. In the AWS console, open Billing and **Cost Management** -> **Data Exports**, and click **Create**, and select **Standard Data Export** and provide the **Export Name**



2. Fill in the **Data table content settings** section as shown below:

- **Include resource IDs:** true
- **Split cost allocation data:** true
- **Time granularity:** Hourly(recommended), daily - possible



3. In the Column selection, select all tables:

▼ **Column selection** (113/113)
Select the columns that you want to include in your data query. This allows you to create exports with consistent schemas, remove sensitive cost data, and reduce the file size of the export. By default, all columns are selected. [Learn more](#)

Filter by column name

☒	Column name ▲	Data type ▼	Description ▼
☒	bill_bill_type	String	The type of bill that this report covers.
☒	bill_billing_entity	String	Helps you identify whether your invoices or transactions are for AWS Marketplace or for purchases of other AWS services.
☒	bill_billing_period_end_date	Timestamp	The end date of the billing period that is covered by this report, in UTC. The format is YYYY-MM-DDTHH:mm:ssZ.
☒	bill_billing_period_start_date	Timestamp	The start date of the billing period that is covered by this report, in UTC. The format is YYYY-MM-DDTHH:mm:ssZ.
☒	bill_invoice_id	String	The ID associated with a specific line item. Until the report is final, the InvoiceId is blank.
☒	bill_invoicing_entity	String	The AWS entity that issues the invoice.
☒	bill_payer_account_id	String	The account ID of the paying account. For an organization in AWS Organizations, this is the account ID of the management account.
☒	bill_payer_account_name	String	The account name of the paying account. For an organization in AWS Organizations, this is the name of the management account.

4. Fill in the **Data export delivery options** section as shown below:

- **Compression type and file format:** Parquet – Parquet
- **File versioning:** Overwrite existing data export file

Data export delivery options
Define your delivery options including file format and overwrite rules. The data export refresh cadence cannot be changed at this time.

Data export refresh cadence

Daily - Export is refreshed at least one time per day

Compression type and file format

Choose the compression type and file format for your export.

☒ Parquet - Parquet

☐ gzip - text/csv

File versioning

Choose whether you want each version of the data export file to overwrite the previous version or to be delivered in addition to the previous versions.

☒ Overwrite existing data export file

Overwriting exports can save on Amazon S3 storage costs.

☐ Create new data export file

Creating new export versions allows you to track the changes in cost and usage data over time.

5. Within the **Data export storage settings** section:

- click **Configure** for the **S3 bucket** section, and fill the opened form with the following values:
 - Choose **Create a bucket** (recommended)
 - Set **S3 bucket name**
 - **Region:** the same with glue

Data export storage settings
Enter a destination in Amazon S3 where your data export will be stored. Amazon S3 is object storage built to store and retrieve data.

S3 bucket
[Empty text field] Configure

S3 path prefix
Enter the S3 path prefix that you want prepended to the name of your data export.
[MyS3PathPrefix]



Configure S3 bucket

Create a bucket or select an existing bucket
Data Exports will deliver your export to the S3 bucket that you specify.

☒ Create a bucket ☐ Select existing bucket

S3 bucket name
[Empty text field]
Bucket name must be globally unique and must not contain spaces or uppercase letters. [Learn more](#)

Region
US East (N. Virginia)

Bucket policy
This policy will be applied with the S3 bucket you specified above.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "EnableAWSDATAExportsToWriteToS3AndCheckPolicy",
      "Effect": "Allow",
      "Principal": {
        "Service": [
          "billingreports.amazonaws.com",
          "bcm-data-exports.amazonaws.com"
        ]
      }
    }
  ]
}
```

Cancel Create bucket

6. Set the **S3 path prefix**, for example, *athena*:

Data export storage settings
Enter a destination in Amazon S3 where your data export will be stored. Amazon S3 is object storage built to store and retrieve data.

S3 bucket
[Empty text field] Configure

S3 path prefix
Enter the S3 path prefix that you want prepended to the name of your data export.
athena

7. Set tags (optionally), and click **Create**.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel Create

2.1.1.2 Configuring report replication

In cases where a main billing account is linked to other accounts, export data should be created in the main billing account.

When the report is transferred to the S3 bucket, you need to configure replication to the required linked accounts, according to the steps below:

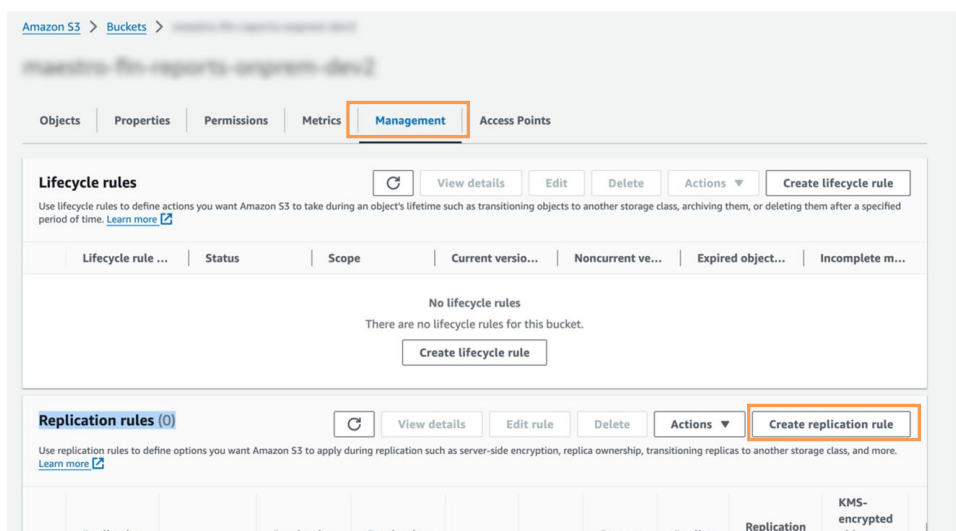
1. Enable versioning, and create the IAM role with the following policy.



Note: in the example below, we use DOC-EXAMPLE-BUCKET as a resource name. Here, put the name of the S3 bucket you created at Step 5 of the Configuring Report section.

```
{
  "Version": "2012-10-17",
  "Id": "",
  "Statement": [
    {
      "Sid": "Set-permissions-for-objects",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::source-bucket-acct-ID:role/service-role/source-acct-IAM-role"
      },
      "Action": ["s3:ReplicateObject", "s3:ReplicateDelete"],
      "Resource": "arn:aws:s3::DOC-EXAMPLE-BUCKET/*"
    },
    {
      "Sid": "Set permissions on bucket",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::source-bucket-acct-ID:role/service-role/source-acct-IAM-role"
      },
      "Action": ["s3:List*", "s3:GetBucketVersioning", "s3:PutBucketVersioning"],
      "Resource": "arn:aws:s3::DOC-EXAMPLE-BUCKET"
    }
  ]
}
```

2. In S3, open the report bucket, and go to **Management -> Replication Rules** and click **Create Replication rule**



3. Set the parameters as follows:

- **Replication rule name**
- **Status:** enabled

Replication rule configuration

Replication rule name

Enter rule ID

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status
Choose whether the rule will be enabled or disabled when created.

☒ Enabled

☐ Disabled

Priority
The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

- **Choose a rule scope:** Apply to all objects in the bucket

Source bucket

Source bucket name

Source Region
Europe (Ireland) eu-west-1

Choose a rule scope

☒ Limit the scope of this rule using one or more filters

☐ Apply to all objects in the bucket

Filter type
You can filter objects by prefix, object tags, or a combination of both.

Prefix
Add a filter to limit the scope of this rule to a single prefix.

Enter prefix

Don't include the bucket name in the prefix. Using certain characters in key names can cause problems with some applications and protocols.

Tags
You can limit the scope of this rule to the key value pairs added below.

Add tag

- **Encryption:** set if required
- **Destination storage class:** Change the storage class for the replicated objects – **false**

Encryption
Server-side encryption protects data at rest.

☐ Replicate objects encrypted with AWS Key Management Service (AWS KMS)
Replicate SSE-KMS and DSE-KMS encrypted objects.

Destination storage class
Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

☐ Change the storage class for the replicated objects

- **Destination:** Specify a bucket in another account
- **Account ID:** Account ID
- **Bucket name:** The name of the bucket to receive replicated objects
- **Destination Region:** should be the same with as the main one

Destination

Destination

You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

- ☐ Choose a bucket in this account
- ☒ Specify a bucket in another account

Account ID

123456789012

Bucket name

Choose the bucket that will receive replicated objects.

replication

Destination Region

Europe (Ireland) eu-west-1

- ☐ Change object ownership to destination bucket owner

Objects in the source bucket not owned by the source bucket owner will be replaced with access policy that grants full permission to the destination bucket owner

- **IAM role:** Choose the role already created at the beginning of this section (Step 1)

IAM role

- ☐ Create new role
- ☒ Choose from existing IAM roles
- ☐ Enter IAM role ARN

IAM role

Choose IAM role



View

- **Additional replication options:** Depending on your needs

Additional replication options

- ☐ Replication Time Control (RTC)

Replication Time Control replicates 99.99% of new objects within 15 minutes and includes replication metrics. Additional fees will apply. [Learn more](#)

- ☐ Replication metrics

With replication metrics, you can monitor the total number and size of objects that are pending replication, and the maximum replication time to the destination Region. You can also view and diagnose replication failures. CloudWatch metrics fees apply. [Learn more](#) or see [Amazon CloudWatch pricing](#)

- ☐ Delete marker replication

Delete markers created by S3 delete operations will be replicated. Delete markers created by lifecycle rules are not replicated. [Learn more](#)

- ☐ Replica modification sync

Replicate metadata changes made to replicas from the destination bucket to the source bucket. [Learn more](#)

Cancel

Save

Click **Save** once all parameters are specified.

4. Go to **Management -> Replication Rules -> Actions:**

[Amazon S3](#) > [Buckets](#) > [mainline-ftp-reports-logs-prod-us-east-2](#)

Objects

Properties

Permissions

Metrics

Management

Access Points

Lifecycle rules

↻

View details

Edit

Delete

Actions ▾

Create lifecycle rule

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Lifecycle rule ...	Status	Scope	Current versio...	Noncurrent ve...	Expired object...	Incomplete m...
No lifecycle rules						
There are no lifecycle rules for this bucket.						
Create lifecycle rule						

Replication rules (0)

↻

View details

Edit rule

Delete

Actions ▾

Create replication rule

Use replication rules to define options you want Amazon S3 to apply during replication such as server-side encryption, replica ownership, transitioning replicas to another storage class, and more. [Learn more](#)

Replication	Destination	Destination	Storage	Replica	Replication	KMS-encrypted objects
-------------	-------------	-------------	---------	---------	-------------	-----------------------

- Set the **Source bucket Account ID** and click **Generate policies**.

Source bucket account settings

Source bucket account ID
To use this bucket as the destination for replicated objects, enter the account ID for the source bucket.

12 digit number

Generate policies

Policies

Bucket policy
If you have write permissions for the bucket, S3 will update the bucket policy automatically when you choose Apply policy.

► **View bucket policy**

Copy **Apply settings**

KMS policy
Before saving KMS encryption settings, ensure that all producers and consumers have access to the specified encryption key. Apply the policy to the KMS key on the KMS console.

► **View KMS policy**

Copy

Close

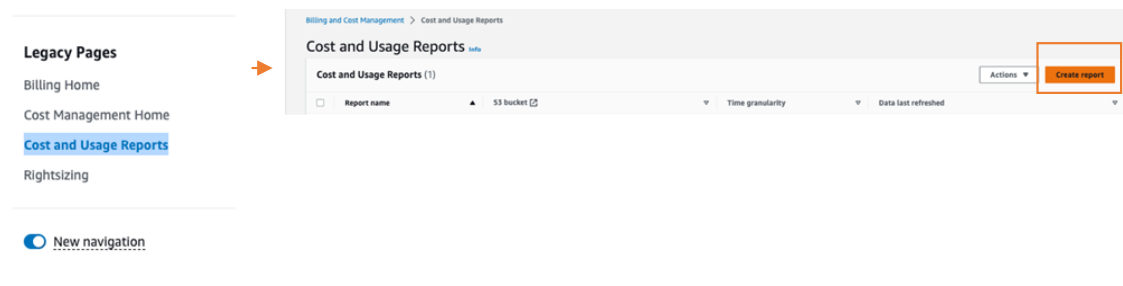
In several hours the data will be copied to the target bucket

2.1.2 Creating a CUR Report

Although we recommend using CUR 2.0 reports, you can still use the previous version.

To configure a CUR report, do this:

1. Go to Billing and Cost Management-> Legacy Pages -> Cost and Usage Reports, and click **Create Report**:



2. At the **Report Content** step, specify the following information and click **Next**:
 - **Report name** ("maestro-cur-hourly" in the example).
 - In **Additional content**, select **Include resource IDs**.
 - In **Data Refresh Settings**, select **Refresh Automatically**.

Report name

Report name - required

Report name must be unique, not include spaces, and contain only alphanumeric and characters ! - _ . * ' ()

Report content

AWS Cost and Usage Reports track your AWS usage and provide estimated charges associated with your account. Each report contains line items for each unique combination of AWS products, usage type, and operation that you use in your AWS account.

Default content

- ☒ Account identifiers
- ☒ Invoice and Bill Information
- ☒ Usage amount and unit
- ☒ Rates and cost
- ☒ Product attributes (instance type, operating system, and region)
- ☒ Pricing attributes (offer types and lease lengths)
- ☒ Reservation identifiers and related details (for Reserved Instances only)

Additional content

☒ **Include resource IDs**
When resources are created, AWS assigns each resource a unique resource ID. Including individual resource IDs in your report can significantly increase the file size.

☐ **Split cost allocation data**
Include detailed cost and usage for shared resources (only available for Amazon ECS). These resources are primarily used for cost allocation and aren't available in billing or invoice reports. Including these resources introduces new rows and columns in the Cost and Usage Report and might increase file size.

Report data processing settings

Data refresh settings

Choose if you want Cost and Usage Reports to refresh if AWS applies refunds, credits, or support fees to your account after finalizing your bill.

☒ **Refresh automatically**
AWS updates the report in your bucket at least once a day. When a report refreshes, a new report is uploaded to Amazon S3.

Cancel

Next

- At the **Delivery options** step, specify the following delivery options and click **Next**:
 - Set **S3 path prefix** to **athena**
 - Set **Report data time granularity** to **Hourly**
 - Set **Report data integration** to **Amazon Athena**

Set delivery options

Configure S3 Bucket

Configure S3 Bucket

Configure

Report delivery options

S3 path prefix - required

athena

Report data time granularity

Choose the time granularity for how you want the line items in the report to be aggregated.

☒ Hourly

☐ Daily

☐ Monthly

Report versioning

Choose whether you want each version of the report to overwrite the previous version of the report or to be delivered in addition to the previous versions.

☐ Create new report version

Delivering new report versions can improve audibility of billing data over time.

☒ Overwrite existing report

Overwriting reports can save on Amazon S3 storage costs.

Report data integration

☒ Amazon Athena

☐ Amazon Redshift

☐ Amazon QuickSight

Compression type

Parquet

File format: Parquet

4. Configure an S3 bucket.

In the pop-up window, you can either select an existing bucket or create a new one. In our example, we created a new bucket named in Frankfurt region.

Set delivery options

Configure S3 Bucket

Configure S3 Bucket

Configure

Report delivery options

S3 path prefix - required

athena

Report data time granularity

Choose the time granularity for how you want the line items in the report to be aggregated.

☒ Hourly

☐ Daily

☐ Monthly

Report versioning

Choose whether you want each version of the report to overwrite the previous version of the report or to be delivered in addition to the previous versions.

☐ Create new report version

Delivering new report versions can improve audibility of billing data over time.

☒ Overwrite existing report

Overwriting reports can save on Amazon S3 storage costs.

Report data integration

☒ Amazon Athena

☐ Amazon Redshift

☐ Amazon QuickSight

Compression type

Parquet

File format: Parquet

Configure S3 Bucket

Create a bucket or Select existing bucket

☒ Create a bucket

☐ Select existing bucket

Create new bucket

S3 bucket name

Region

Europe (Ireland)

```
{
  "Version": "2008-10-17",
  "Id": "Policy1335892530063",
  "Statement": [
    {
      "Sid": "Stmt1335892150622",
      "Effect": "Allow",
      "Principal": {
        "Service": "billingreports.amazonaws.com"
      },
      "Action": [
        "s3:GetBucketAct",
        "s3:GetBucketPolicy"
      ],
      "Resource": "arn:aws:s3:::maestro-fin-reports-onprem-dev2"
    }
  ]
}
```

☒ The following default policy will be applied to your bucket

Cancel Save



- In case you create a new bucket, please note that the bucket name should be unique within AWS across all the existing accounts. Otherwise, bucket creation will fail.
- You will also need to review the default bucket policy and approve it.

- All resources used for gathering the billing information should be hosted in one region.

Save the cost and usage report bucket name for further referencing and sharing.

5. Verify the settings in the Review and Create step:

The report configuration will take about 24 hours.

When the report is ready, you can find it in **My Billing Dashboard** → **Cost & Usage reports**

Cost and Usage Reports (1)				
	Report name	S3 bucket	Time granularity	Data last refreshed
☐	maestro-cur-hourly	athena/maestro-cur-hourly/date-range/	Hourly	February 21, 2024, 11:42 (UTC+01:00)

2.2 CONFIGURING AWS ATHENA SERVICE

AWS Athena Service allows you to query the billing data for your account. Service configuration procedure includes three steps:

1. Creating a bucket for AWS Athena query results.
2. Configuring an AWS Athena service.
3. Configuring a AWS Athena query.

2.2.1 Configuring AWS Athena for CUR 2.0

2.2.1.1 Configuring AWS Athena (CUR 2.0 Crawler)

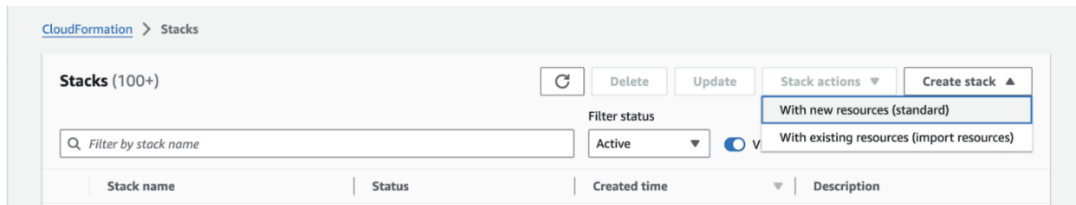
AWS Athena is an SQL-based query service used to retrieve data retrieving from AWS S3 quickly and easily. The queries set up by the service are used to retrieve the necessary billing information for further usage by Maestro.

To configure AWS Crawlers we are recommended to use CloudFormation.

You can find the necessary CloudFormation template in additional documents: **crawler_hourly.yaml** and **crawler_daily.yaml**

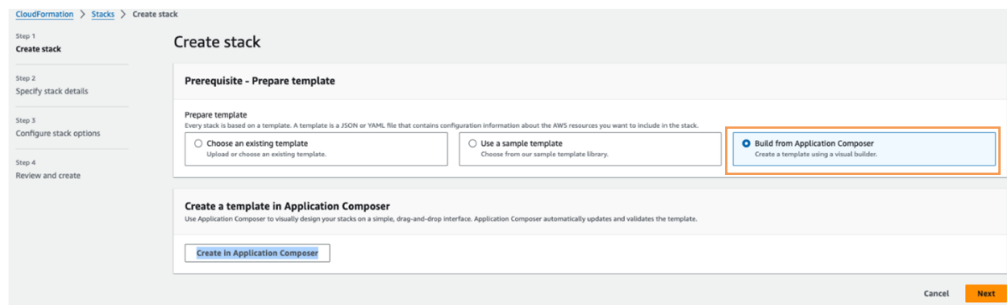
To run the template, do the following:

1. In AWS CloudFormation, click **“Create stack”** → **With new resources** (standard)

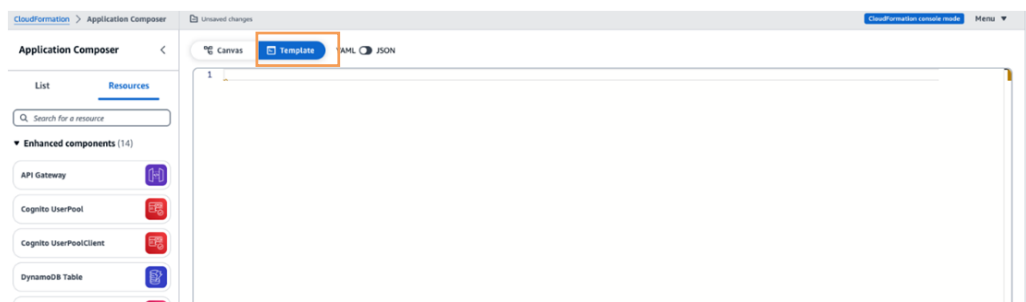


2. Complete Prerequisite – Prepare a template:

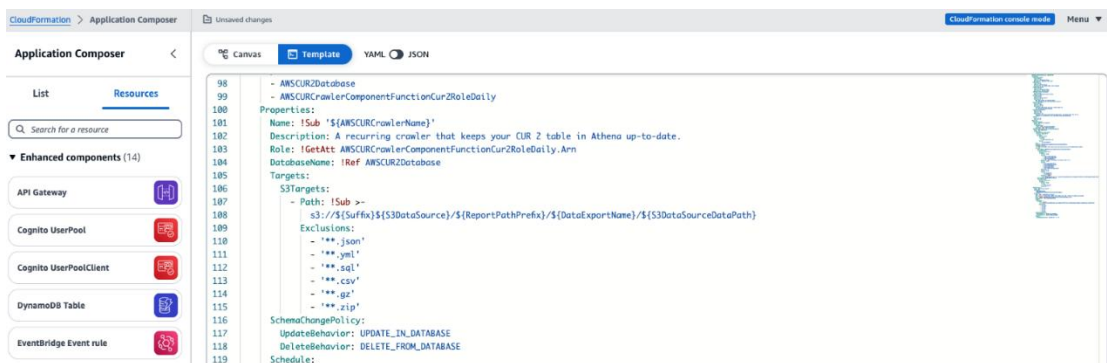
- Create a template in Application Composer: [Create in Application Composer](#)



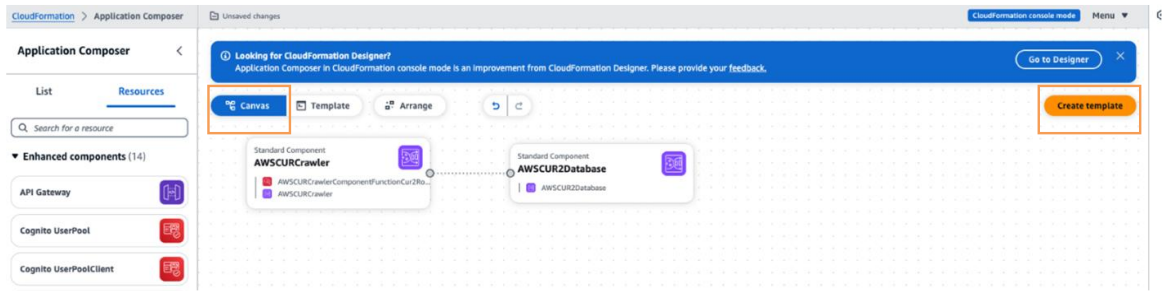
- Choose **Template** in the designer



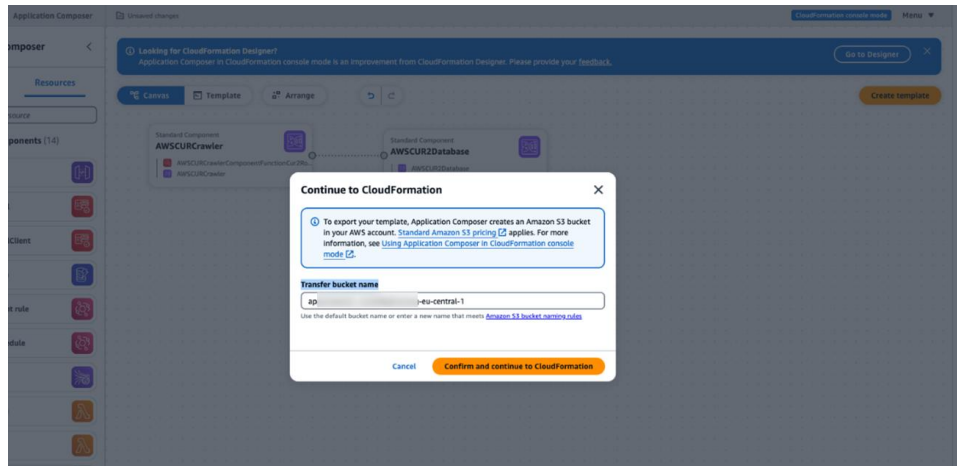
- Insert the **crawler_hourly.yaml** or **crawler_daily.yaml** template (depending on the granularity you prefer)



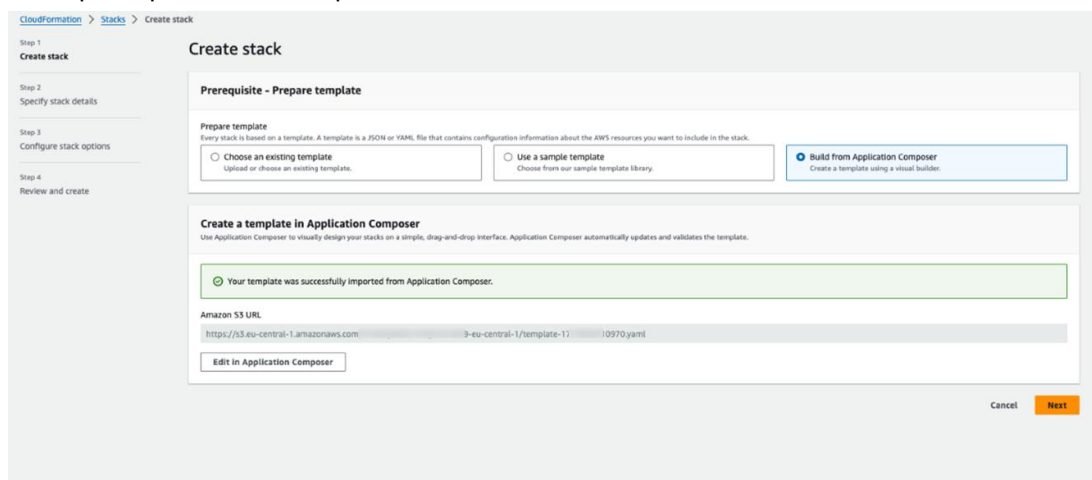
- Go back to the **Canvas** and click **create template**



e. Set the Transfer bucket name and then **“Confirm and continue to CloudFormation”**



3. Once prerequisites are completed, click **Next**.



4. Specify stack details and click **Next**:

- **Stack name** : <STACK_NAME>
- **AWSCURCrawlerName**: <AWS_CUR2_CRAWLER_NAME>
- **DataExportName**: <DATA_EXPORT_NAME> - the name of the data export which was created in the “CREATING AN EXPORT DATA” step
- **GlueDatabaseName**: <GLUE_DATABASE_NAME> recommended to use similar name to “DataExportName”
- **ReportPathPrefix**: <REPORT_PATH_PREFIX> path from “CREATING AN EXPORT DATA” Data export storage settings section (in our example – *athena*)
- **S3DataSource**: <S3_DATA_SOURCE> s3 bucket where store reports, has been set in Data export storage settings section

- **S3DataSourceDataPath:**<S3_DATA_SOURCE_DATA_PATH>: by default data
- **Suffix:**<SUFFIX> - set environment suffix, for example, *dev-*

Provide a stack name

Stack name

maestro-daily

Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

AWSCURCrawlerName
Crawler Name

maestro-daily

DataExportName
Name data export ex maestro-daily s3:/ /athena/maestro-daily/

maestro-daily

GlueDatabaseName
database name glue

maestro-daily

ReportPathPrefix
directory name which was set in Data Export ex s3:/ /athena

athena

S3DataSource
s3 bucket where store reports

S3DataSourceDataPath
directory name which generated automatically, by default data but can be changed for ex data223 ex s3:/ /athena/data

data

Suffix

dev-

Cancel Previous Next

5. Review parameters:

CloudFormation > Stacks > athena-billing-setup-onprem-cur2-v2 > Update stack

Step 1
Update stack

Step 2
Specify stack details

Step 3
Configure stack options

Step 4
Review athena-billing-setup-onprem-cur2-v2

Review athena-billing-setup-onprem-cur2-v2

Step 1: Specify template

Template

Template URL
Use current template

Stack description
-

Step 2: Specify stack details

Parameters (8)

Search

< 1 > ⌵

Key	Value
-----	-------

- Set **true** at “I acknowledge that AWS CloudFormation might create IAM resources” and click **Submit**.

Capabilities

The following resource(s) require capabilities: [AWS::IAM::Role]

This template contains Identity and Access Management (IAM) resources that might provide entities access to make changes to your AWS account. Check that you want to create each of these resources and that they have the minimum required permissions. [Learn more](#)

☐ I acknowledge that AWS CloudFormation might create IAM resources.

View change setCancelPreviousSubmit

2.2.1.2 Creating a Bucket for AWS Athena Query Results

To create a new bucket, perform the steps given below.

- In the AWS S3 service page, click **Create bucket** and set up a new bucket parameters, keeping to the following rules:
 - The bucket should be created in the same region for which the Cost and Usage report was set up.
 - The bucket name should be unique within AWS across all the existing accounts.

Create bucket wizard may look different for your account, but its content will be the same.

Amazon S3 > Buckets > Create bucket

Create bucket info

Buckets are containers for data stored in S3. [Learn more](#)

General configuration

AWS Region
Europe (Ireland) eu-west-1

Bucket name info
f

Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix

Object Ownership info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

- At the **Set permissions** step, make sure that **Block all public access** is selected:

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block Public Access settings for this account are currently turned on
[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☒ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☒ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☒ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☒ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

3. At the **Configure options** step, select **Default encryption** as shown in the screenshot (also verify that other values are set as shown, which is most likely to be default values):

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☒ **Disable**
☐ Enable

Tags - optional (0)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

No tags associated with this bucket.

Add tag

Default encryption Info

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type Info

☒ **Server-side encryption with Amazon S3 managed keys (SSE-S3)**
☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

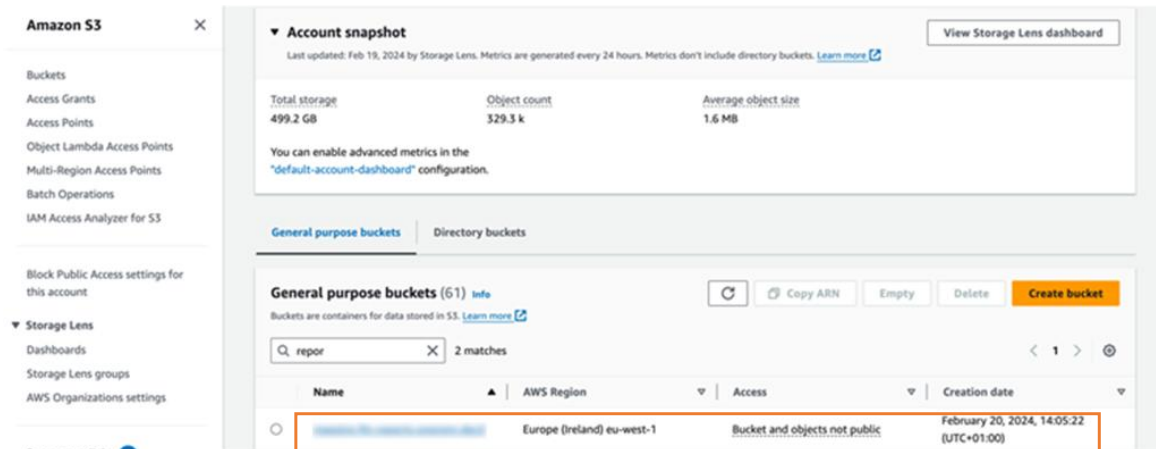
☐ Disable
☒ **Enable**

2.2.2 Configuring AWS Athena for CUR

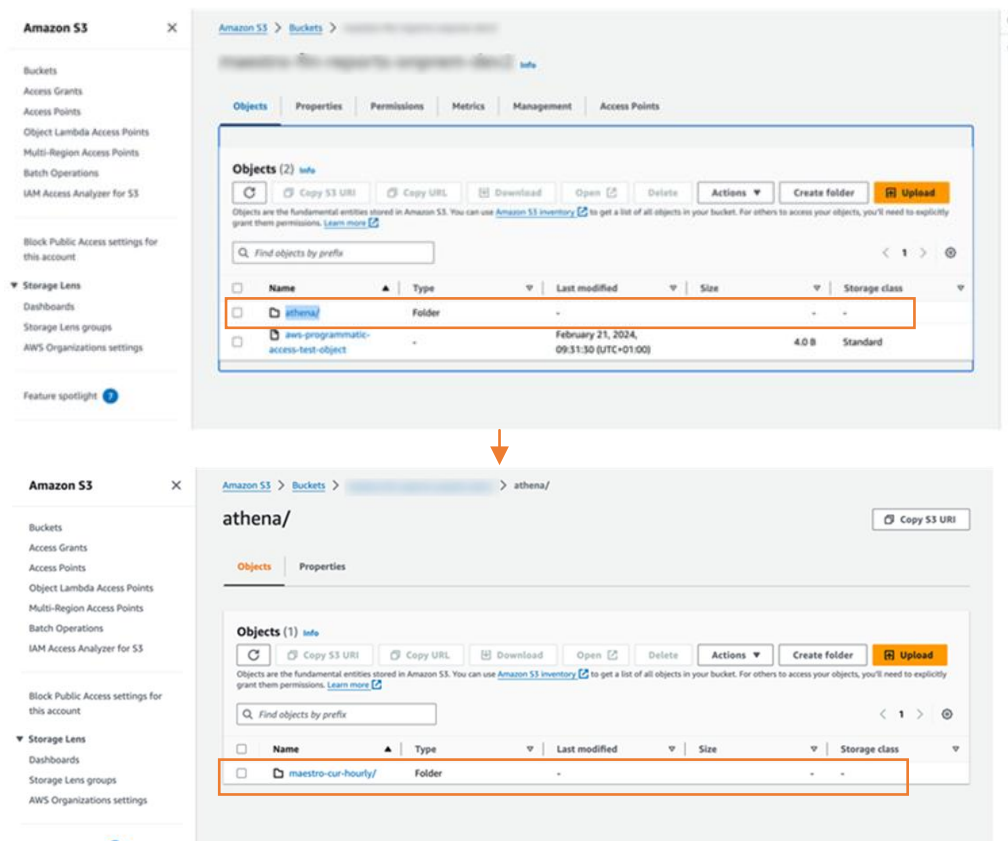
2.2.2.1 Configuring AWS Athena (CUR Crawler and IAM roles)

If you decided to use CUR type reports, follow the steps below to configure the Athena:

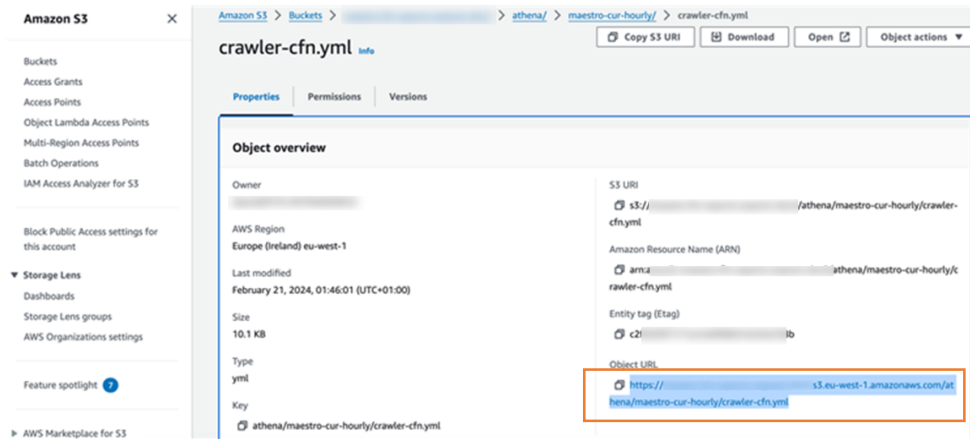
1. At the AWS S3 service page, find the AWS S3 bucket you created for Cost & Usage report:



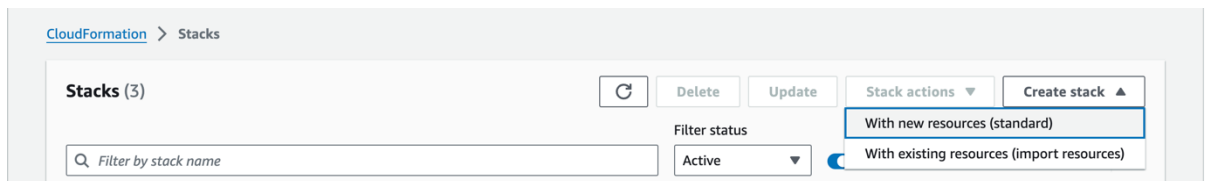
2. In the bucket, go to **Athena** → the **<report_name>** folder:



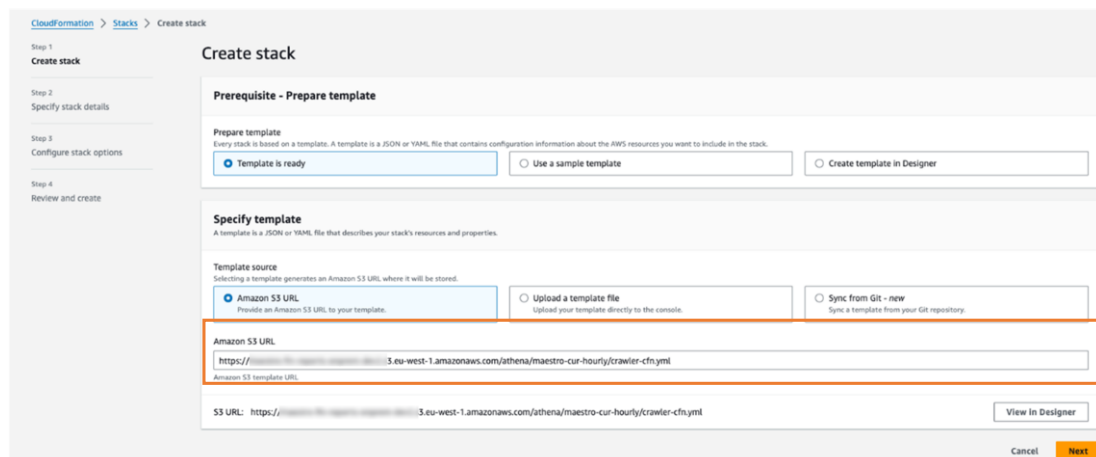
3. In the report folder, find the AWS CloudFormation template named **“crawler-cfn.yml”**. This template is pre-configured by AWS to enable the automatic Athena setup.
Copy the template’s URL for further usage:



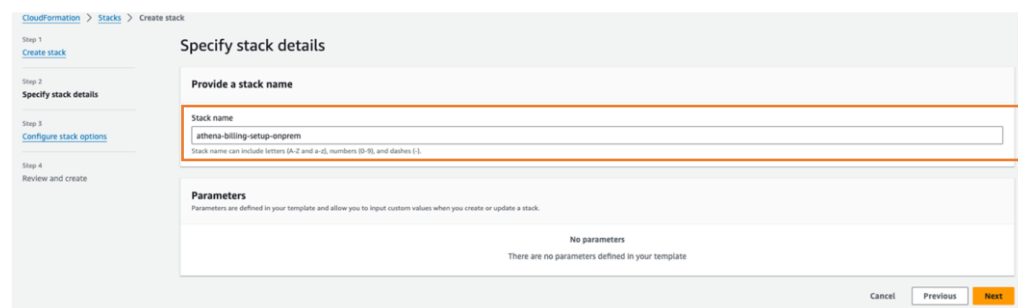
- Go to the **CloudFormation** service page → **Stacks**, and click **Create Stack-> with new resources**:



- At the **Specify Template** step, paste the URL of the CloudFormation template, and click **Next**:



- At the **Specify stack details** step, specify the stack name and click **Next**. In our example, we use “athena-billing-setup-onprem” as the stack name:



- At the **Configure stack options** step, review the details and click Next. No updates are needed:

IMPORTANT NOTE:

The selected User or role need to have permissions create IAM roles, policies, glue, lambdas.

- In the **Review** page, review the details and check the box **“I acknowledge that AWS CloudFormation might create IAM resources”**. After that, click **Create stack**.

- In the **Stack Details** section, check that the stack was created (this may take several minutes):

- Switch to the **Resources** tab and find the physical ID of the **AWSCURDatabase**. The resource type is **AWS:Glue:Database**.

Delete Update Stack actions ▼ Create stack ▼			
Stack info Events Resources Outputs Parameters Template Change sets			
Resources (11)			
Search resources			
Logical ID	Physical ID	Type	Status
AWSCURCrawler	AWSCURCrawler-maestro-cur-hourly	AWS::Glue::Crawler	CREATE_COMPLETE
AWSCURCrawlerComponentFunction	athena-billing-setup-onpr-AWSCURCrawlerComponentFunction-nlM56mrRl0cG	AWS::IAM::Role	CREATE_COMPLETE
AWSCURCrawlerLambdaExecutor	athena-billing-setup-onpr-AWSCURCrawlerLambdaExecutor-ZAbmEtc3zJqG	AWS::IAM::Role	CREATE_COMPLETE
AWSCURDatabase	athenacurfn-maestro-cur-hourly	AWS::Glue::Database	CREATE_COMPLETE

Save the **AWSCURDatabase ID** for further usage. It will be needed for retrieving <glue_athena_main_database_name> and <glue_athena_main_table_name> .

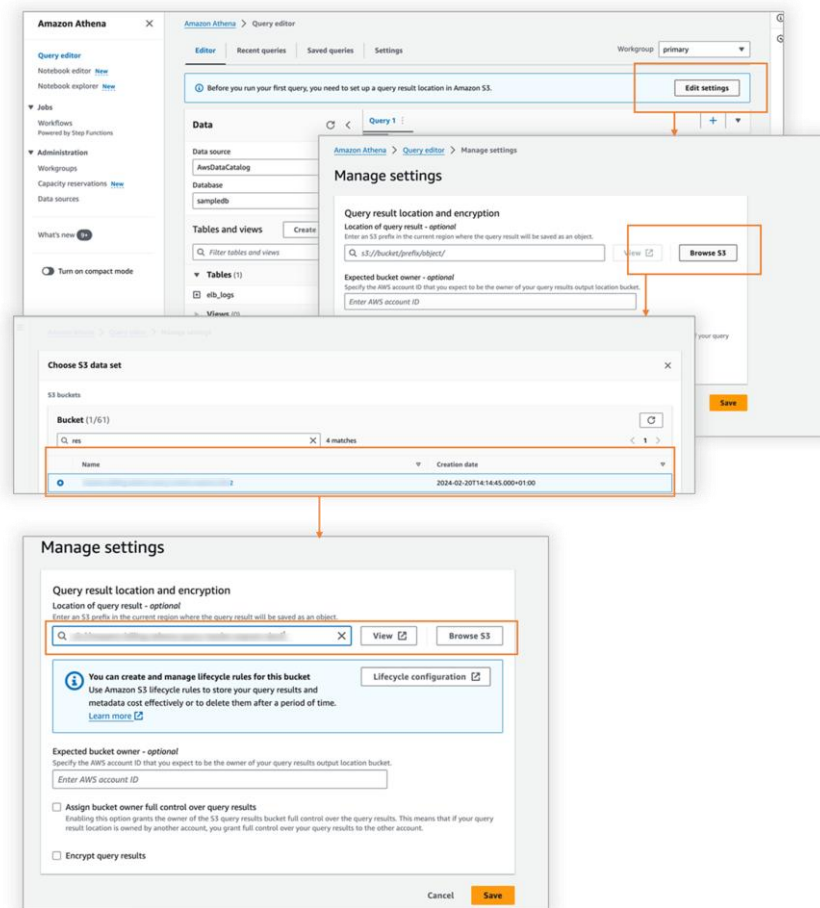
2.2.2.2 Configuring Athena Query for CUR

Once AWS Athena is activated, you can configure the query to retrieve the billing data properly. To do it, follow these steps:

1. Go to the **AWS Athena service page** and open **Query editor**:

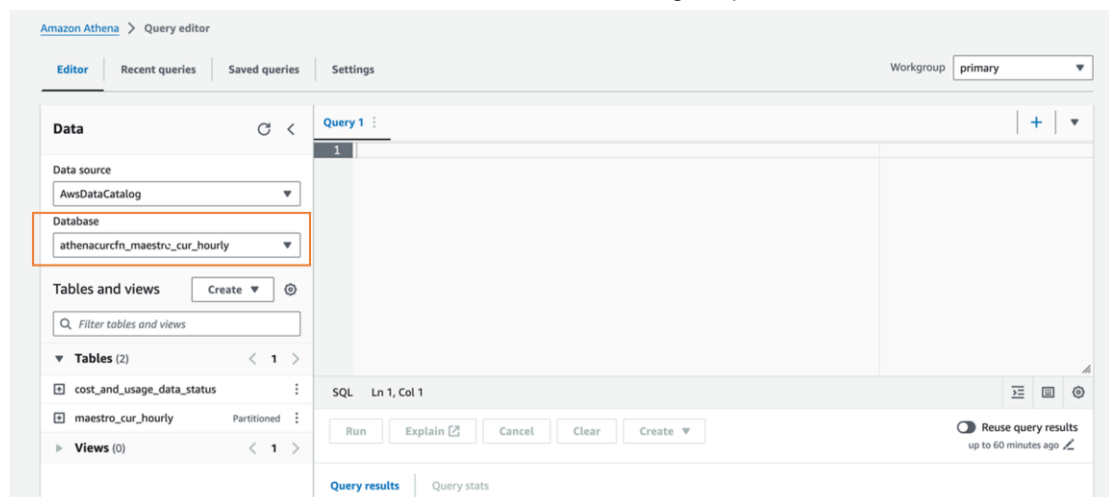
The screenshot shows the Amazon Athena console interface. On the left is a navigation sidebar with options like 'Query editor', 'Notebook editor', and 'Administration'. The main content area has a dark header with 'Amazon Athena' and 'Start querying data instantly.' Below this, there's a 'Get started' section with two radio buttons: 'Query your data with Trino SQL' (selected) and 'Analyze your data using PySpark and Spark SQL'. The 'Launch query editor' button is highlighted with an orange box. To the right, a 'Pricing' section shows costs for SQL queries and Spark calculations. At the bottom, a 'How it works' section shows a diagram of a bucket and the step '1. Point to your data source'.

2. Select the query result location bucket. Use the one created at the Creating a Bucket for AWS Athena Query Results step:



Review, click **Save** and **Next**.

3. Check that the necessary database and tables are used.
The database name includes the name of the Cost and Usage report created here:

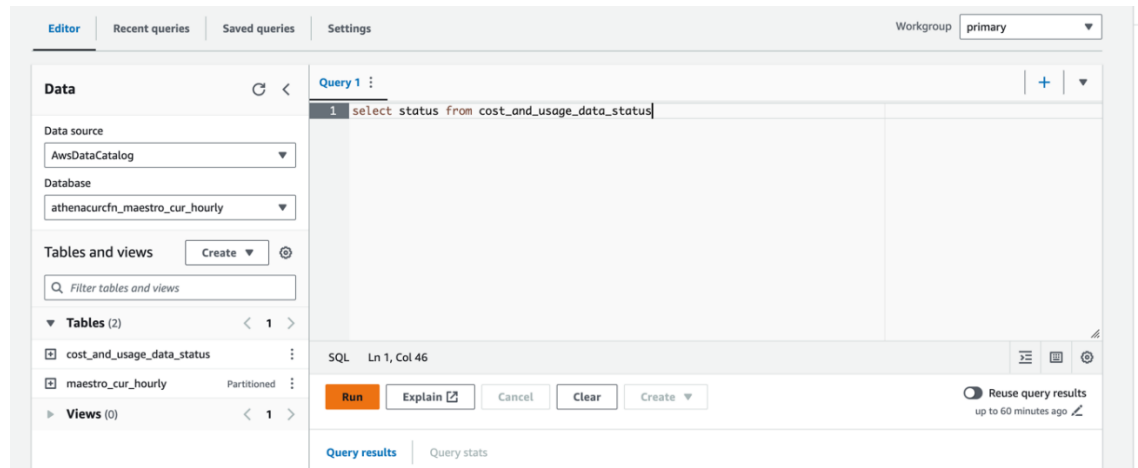


- 3.1. Input the new query. The query text is:

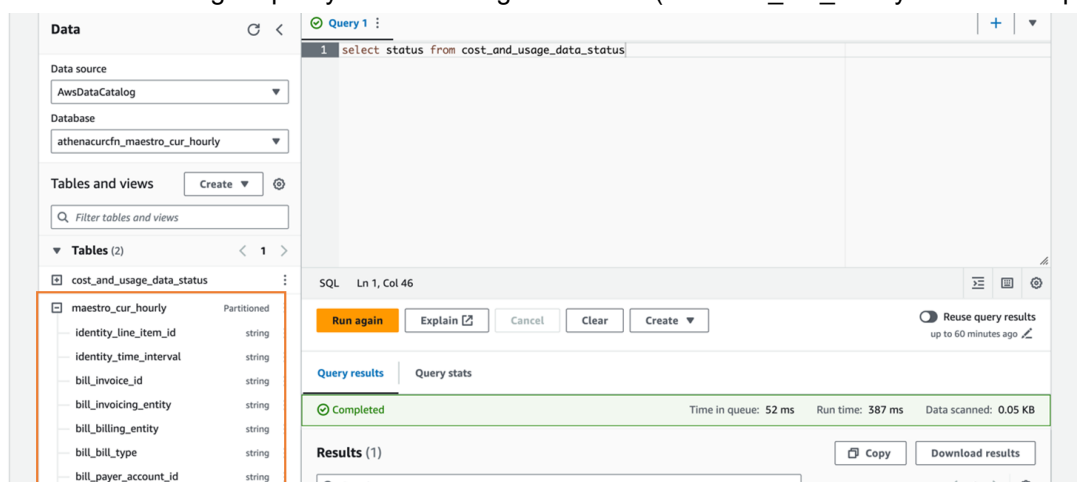
```
status from cost_and_usage_data_status
```

- 3.2. Click **Run Query**

- 3.3. Check that the Query was completed:



3.4. In several minutes, a new table with the billing data will appear, and its name will be the name of the Cost and Usage report you have configured earlier (“maestro_cur_hourly” in our example):



2.3 CREATING A USER FOR BILLING ACCESS

To provide the Maestro team with all the necessary permissions to query your billing information, you need to create an IAM role and configure it properly.

This is done in two steps:

1. Create an IAM policy.
2. Create an IAM role using this policy.

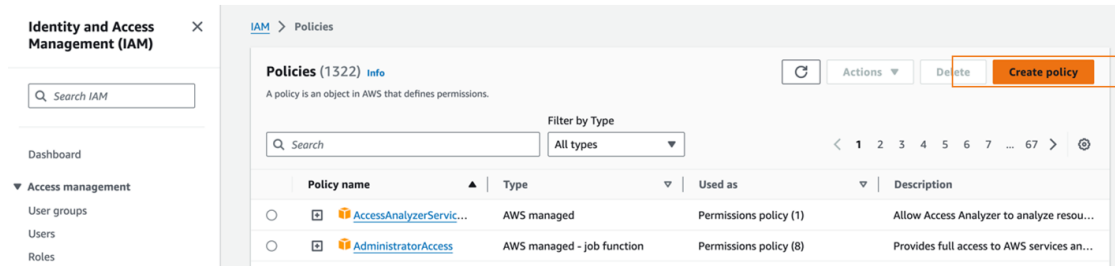
2.3.1 Creating an IAM Policy

You can find the necessary policy in annexes:

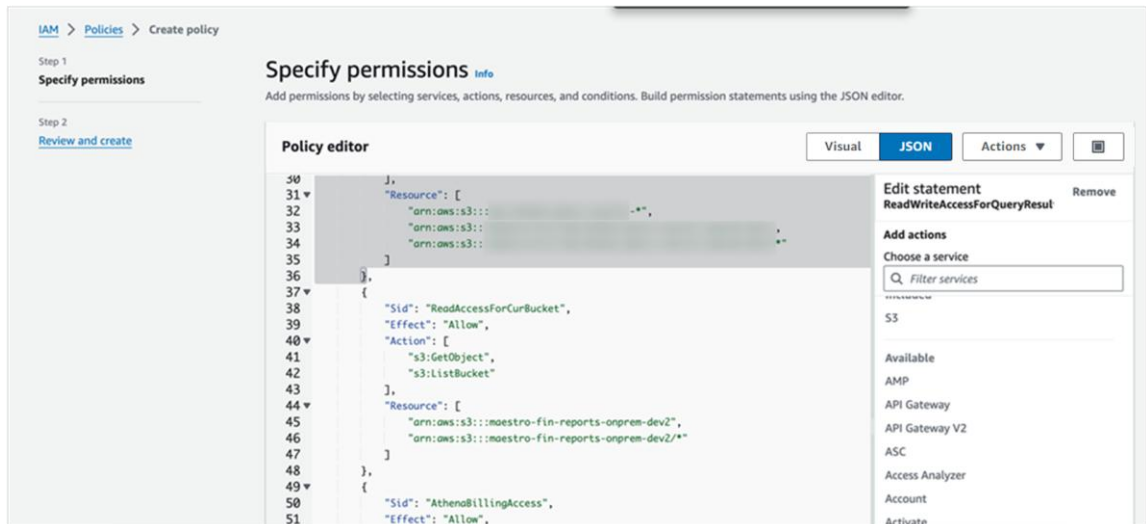
- Cur: Annex A
- Cur2: Annex B

Creating an IAM policy is the first step in a new role configuration. To do it, perform these steps:

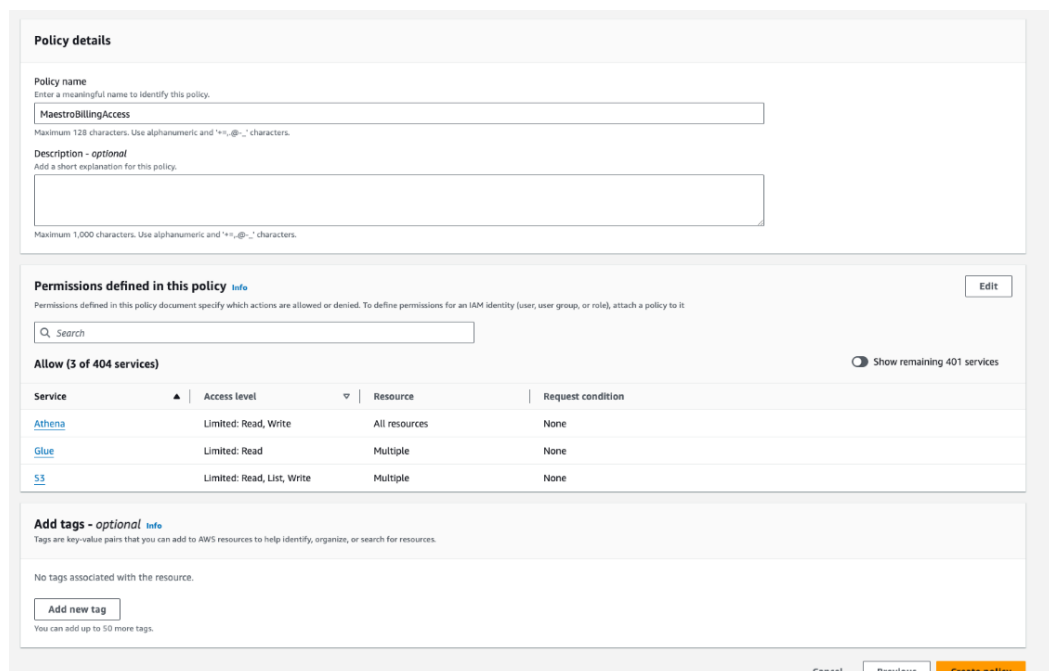
1. In the **Identity and Access Management** service page, got to **Policies** and click **Create policy**:



2. In the **Create Policy** wizard, select the **Json** tab. Insert the IAM policy for billing data access given in Annex to the JSON policy editor, and click **Review Policy**:



3. At the Review Policy step, specify the name of the new policy. In our example, it is **MaestroBillingAccess**. After that, click **Create policy**:



Remember the name of the policy for further usage and sharing.

2.3.2 Creating an IAM User

An IAM user is needed to enable the access to your account's billing data for Maestro with the permissions listed in the newly created policy.

1. In the **Identity and Access Management > IAM users -> Create user**, set User name and click next:

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Specify user details

User details

User name

Maestro3BillingAccessUser

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☐ Provide user access to the AWS Management Console - *optional*
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

i If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel **Next**

2. Choose **Attach policies directly** and find the policy created on the previous step.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1326)

Choose one or more policies to attach to your new user.

Filter by Type

MaestroBillingAccess All types 1 match

☐	Policy name	Type	Attached entities
☐	MaestroBillingAccess	Customer managed	0

3. Review and create a new user:

The screenshot shows the 'Review and create' step of the AWS IAM 'Create user' process. The left sidebar indicates the current step is 'Review and create'. The main content area is titled 'Review and create' and includes a sub-header 'Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.' The 'User details' section shows the user name 'Maestro3BillingAccessUser', console password type 'None', and 'Require password reset' set to 'No'. The 'Permissions summary' section shows a table with columns 'Name', 'Type', and 'Used as', and a message 'No resources'. The 'Tags - optional' section shows 'No tags associated with the resource.' and an 'Add new tag' button. At the bottom right, the 'Create user' button is highlighted in orange.

4. Open the new user, then **Create access key: Security credentials** -> **Create access key**:

The screenshot shows the 'Security credentials' page for a user in the AWS IAM console. The top navigation bar includes 'Permissions', 'Groups', 'Tags (2)', 'Security credentials' (highlighted), and 'Access Advisor'. The 'Console sign-in' section has an 'Enable console access' button. The 'Multi-factor authentication (MFA) (0)' section has buttons for 'Remove', 'Resync', and 'Assign MFA device'. The 'Access keys (1)' section has a 'Create access key' button highlighted in orange.

2.4 ADDITIONAL STEPS

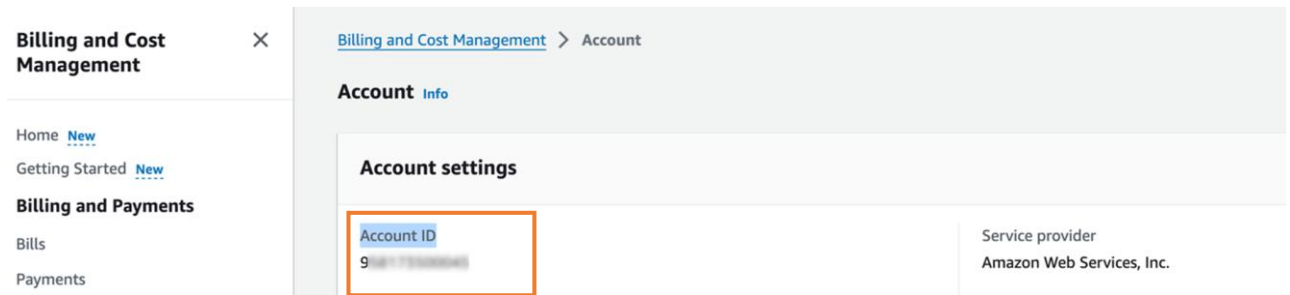
In addition to the steps described before, please also find and remember:

- AWS Account ID
- Cost and Usage reports status file S3 path.

The instructions are given below in this section.

2.4.1 Finding AWS Account ID

AWS Account ID is needed for further referencing when connecting Maestro to your AWS Account. AWS Account ID can be found on the **My Account** page of the **Billing** section.



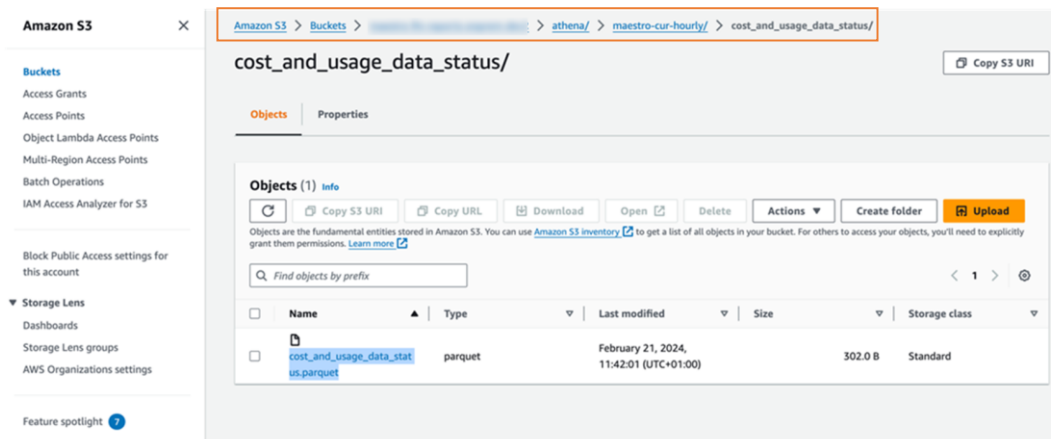
Save this value for further usage and sharing.

2.4.2 Finding Cost and Usage reports status file S3 path (for CUR only)

To find this path, follow these steps:

1. Go to **AWS S3 service page**, and consequently select:

[the cost and usage report bucket] → **athena** → [cost and usage report folder] →
→ **cost_and_usage_data_status** → **cost_and_usage_data_status.parquet**



2. In the file details, find and copy the **Key** value.

cost_and_usage_data_status.parquet [Info](#)

[Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

[Properties](#) [Permissions](#) [Versions](#)

Object overview

Owner	S 2
AWS Region	Europe (Ireland) eu-west-1
Last modified	February 21, 2024, 11:42:01 (UTC+01:00)
Size	302.0 B
Type	parquet
Key	athena/maestro-cur-hourly/cost_and_usage_data_status/cost_and_usage_data_status.parquet

S3 URI
[s3://maestro-reports-us-east-1/athena/maestro-cur-hourly/cost_and_usage_data_status/cost_and_usage_data_status.parquet](#)

Amazon Resource Name (ARN)
[arn:aws:s3:::maestro-reports-us-east-1/athena/maestro-cur-hourly/cost_and_usage_data_status/cost_and_usage_data_status.parquet](#)

Entity tag (Etag)
[9d1a8](#)

Object URL
[https://maestro-reports-us-east-1.s3.eu-west-1.amazonaws.com/athena/maestro-cur-hourly/cost_and_usage_data_status/cost_and_usage_data_status.parquet](#)

Save the Key value for further usage and sharing as the Cost and Usage report status file path.

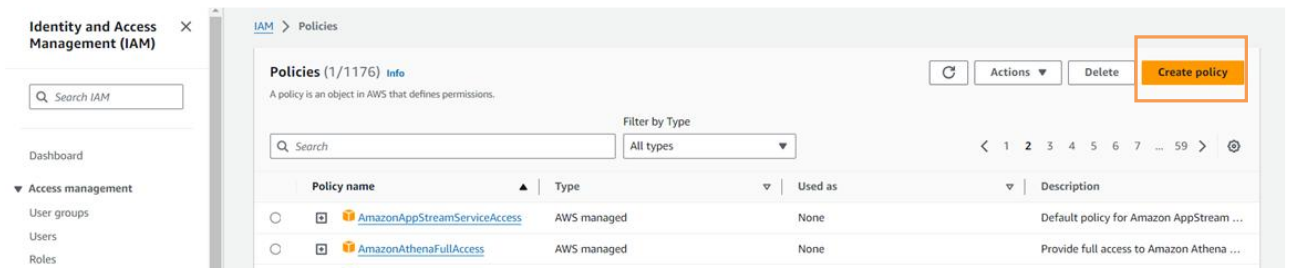
3 CONFIGURING AN AWS ACCOUNT FOR MANAGEMENT

Configuring an account to allow Maestro to manage it, includes two main steps: creating a necessary IAM policy and user, and generating programmatic access credentials.

3.1 CREATING AN IAM POLICY

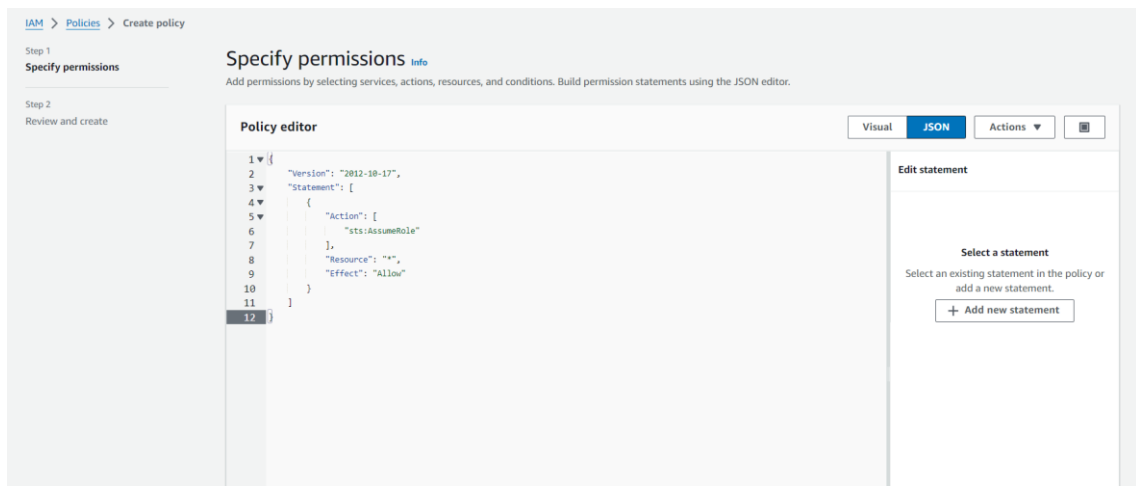
To create a policy for allowing management, do the following:

1. In the **Identity and Access Management** service page, got to **Policies** and click **Create policy**:



2. In the **Create Policy** wizard, select the **Json** tab. Replace the example policy with following text, and click **Review Policy** at the bottom of the page:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "sts:AssumeRole"
      ],
      "Resource": "*",
      "Effect": "Allow"
    }
  ]
}
```



- At the Review Policy step, specify the name of the new policy. In our example, it is **maestro-service-sts-policy**, and click **Create Policy**:

Policy name
Enter a meaningful name to identify this policy.
maestro-service-sts-policy
Maximum 128 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description - optional
Add a short explanation for this policy.
Maximum 1,000 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Permissions defined in this policy [Info](#) [Edit](#)
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Q Search

Allow (1 of 409 services) [Show remaining 408 services](#)

Service	Access level	Resource	Request condition
STS	Limited: Write	All resources	None

Add tags - optional [Info](#)
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

Cancel Previous **Create policy**

3.2 CREATING AN IAM USER

To create IAM user for Maestro on-premise, do the following:

- In the **Identity and Access Management > IAM users -> Create user**, set User name (for example, **maestro-service**) and click **Next**:

IAM > Users > Create user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Specify user details

User details

User name
maestro-service
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, ., @, _ (hyphen)

☐ Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

[Info](#) If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel **Next**

- Choose **Attach policies directly**, find **PowerUserAccess** policy and check the checkbox:

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1180)

Choose one or more policies to attach to your new user.

Search: Filter by Type: 1 match

☒	Policy name	Type	Attached entities
☒	PowerUserAccess	AWS managed - job function	0

► Set permissions boundary - optional

Cancel Previous **Next**

- Find the policy created before (the example "maestro-service-sts-policy") and check the checkbox

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (2/1180)

Choose one or more policies to attach to your new user.

Search: Filter by Type: 1 match

☒	Policy name	Type	Attached entities
☒	maestro-service-sts-policy	Customer managed	0

► Set permissions boundary - optional

Cancel Previous **Next**

- Review and create a new user:

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name: Console password type: Require password reset:

Permissions summary

Name	Type	Used as
maestro-service-sts-policy	Customer managed	Permissions policy
PowerUserAccess	AWS managed - job function	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

You can add up to 50 more tags.

Cancel Previous **Create user**

3.3 GENERATING PROGRAMMATIC ACCESS CREDENTIALS

To generate programmatic access credentials, do the following:

- In the **Identity and Access Management** > **IAM users** -> **Create user** -> **Security Credentials**

IAM > Users > maestro-service

maestro-service [Info](#) Delete

Summary

ARN arn:aws:iam::[redacted]:user/maestro-service	Console access Disabled	Access key 1 Create access key
Created February 21, 2024, 14:02 (UTC+02:00)	Last console sign-in -	

Permissions | Groups | Tags | **Security credentials** | Access Advisor

2. Scroll down to **Access Keys** section -> **Create access key**:

Access keys (1) Create access key

Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. [Learn more](#)

3. Choose the **Other** option:

Step 1
Access key best practices & alternatives

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

- ☐ **Command Line Interface (CLI)**
You plan to use this access key to enable the AWS CLI to access your AWS account.
- ☐ **Local code**
You plan to use this access key to enable application code in a local development environment to access your AWS account.
- ☐ **Application running on an AWS compute service**
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
- ☐ **Third-party service**
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.
- ☐ **Application running outside AWS**
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.
- ☒ **Other**
Your use case is not listed here.

4. Set the “maestro” tag and click **Create access key**:

Step 1
[Access key best practices & alternatives](#)

Step 2 - optional
Set description tag

Step 3
Retrieve access keys

Set description tag - optional [Info](#)

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value
Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

maestro

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ : / = + - @

Cancel Previous **Create access key**

When the key is created, it can be downloaded as displayed below:

Retrieve access keys [Info](#)

Access key

If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key

Secret access key

 AKIATI

 ***** [Show](#)

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

[Download .csv file](#)

[Done](#)

Save the .csv file with the secrets for further sharing with the Maestro team.

ANNEX A – IAM POLICY FOR BILLING DATA ACCESS

This policy is to be used at the Creating an IAM policy step.

The policy enables the read access to the billing reports and queries gathered by Athena.

Please note that the policy includes several placeholders, which contents should be replaced by the values you have obtained when completing the previous configuration steps:

Placeholder	Value to be inserted
<athena_region>	Region in which Athena and other billing-related resources are configured (given as region code, for example, eu-central-1)
<account_id>	ID of the account for which billing is set up. 2.4.1 Finding AWS Account ID
<glue_athena_main_database_name>	RETRIEVING <GLUE_ATHENA_MAIN_DATABASE_NAME> N 2
<glue_athena_cur_table_name>	RETRIEVING <GLUE_ATHENA_CUR_TABLE_NAME> N 4
<athena_query_results_bucket>	Name of the S3 bucket used for storing Athena query results (was created at this step).
<cur_bucket_name>	Name of the S3 bucket used for Cost and Usage report storing (2.1 CREATING A COST & USAGE REPORT).

The policy content is given below:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadAccessGlueForBilling", "Effect": "Allow",
      "Action": [
        "glue:GetPartition",
        "glue:GetPartitions",
        "glue:GetTable"
      ],
      "Resource": [
        "arn:aws:glue:<athena_region>:<account_id>:database/<glue_athena_main_database_name>",
        "arn:aws:glue:<athena_region>:<account_id>:table/<glue_athena_main_database_name>/*",
        "arn:aws:glue:<athena_region>:<account_id>:table/<glue_athena_cur_table_name>",
        "arn:aws:glue:<athena_region>:<account_id>:catalog"
      ]
    },
    {
      "Sid": "ReadWriteAccessForQueryResults", "Effect": "Allow",
      "Action": [
        "s3:GetBucketLocation", "s3:GetObject",
        "s3:ListBucket", "s3:ListBucketMultipartUploads", "s3:ListMultipartUploadParts", "s3:AbortMultipartUpload", "s3:CreateBucket",
        "s3:PutObject"
      ],
      "Resource": [
        "arn:aws:s3:::aws-athena-query-results-*", "arn:aws:s3:::<athena_query_results_bucket>",
        "arn:aws:s3:::<athena_query_results_bucket>/*"
      ]
    },
    {
      "Sid": "ReadAccessForCurBucket", "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::<cur_bucket_name>", "arn:aws:s3:::<cur_bucket_name>/*"
      ]
    },
    {
      "Sid": "AthenaBillingAccess", "Effect": "Allow",
      "Action": [
        "athena:StartQueryExecution", "athena:GetQueryExecution", "athena:GetQueryResults"
      ],
      "Resource": "*"
    }
  ]
}
```

ANNEX B – IAM POLICY FOR BILLING DATA ACCESS CUR 2.0

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadAccessGlueForBilling",
      "Effect": "Allow",
      "Action": [
        "glue:GetPartition",
        "glue:GetPartitions",
        "glue:GetTable",
        "glue:GetDatabases",
        "glue:GetDatabase"
      ],
      "Resource": [
        "arn:aws:glue:eu-central-1:123456789001:database/<database_name>",
        "arn:aws:glue:eu-central-1:123456789001:table/<database_name>/**",
        "arn:aws:glue:eu-central-1:123456789001:catalog"
      ]
    },
    {
      "Sid": "ReadWriteAccessForQueryResults",
      "Effect": "Allow",
      "Action": [
        "s3:GetBucketLocation",
        "s3:GetObject",
        "s3:ListBucket",
        "s3:ListBucketMultipartUploads",
        "s3:ListMultipartUploadParts",
        "s3:AbortMultipartUpload",
        "s3:CreateBucket",
        "s3:PutObject"
      ],
      "Resource": [
        "arn:aws:s3:::<athena_query_results>-*",
        "arn:aws:s3:::<athena_query_results>",
        "arn:aws:s3:::<athena_query_results>/*"
      ]
    },
    {
      "Sid": "ReadAccessForCurBucket",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::<s3_bucket_report_name>",
        "arn:aws:s3:::<s3_bucket_report_name>/*"
      ]
    },
    {
      "Sid": "AthenaBillingAccess",
      "Effect": "Allow",
      "Action": [
        "athena:StartQueryExecution",
        "athena:GetQueryExecution",
        "athena:GetQueryResults"
      ],
      "Resource": "*"
    }
  ]
}
```

Placeholder	Value to be inserted
< database_name >	Database name which was created for store billing report. Create after first execute crawler
< athena_query_results >	S3 bucket which was created for store query results from athena
< s3_bucket_report_name >	S3 bucket which was created for store billing reports

VERSION HISTORY

Version	Date	Summary
3.2	February 19, 2026	Reviewed
3.1	April 23, 2024	Added the Management section
3.0	April 10, 2024	Added CUR 2.0 instruction
2.0	July 3, 2020	Combined from two initial docs
1.1	November 1, 2019	Chapter “Register Application in Azure AD” updated Page headers added
1.0	April 16, 2019	Combined from two initial docs